Log Transformations

Growth rates, percent errors, and elasticity

Foundations of Regression and Predictive Modeling

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A straight line is a promise

In simple linear regression, a slope means:

Add this much to y for one more unit of x.

Examples:

- one more year of age: subtract about 0.7 BPM,
- one more square foot: add \$500 to house price.

One-unit change in x, β_1-unit change in y, no matter the value of x.

March 2014

On March 23, 2014, WHO announced an outbreak of Ebola virus disease in Guinea.

Imagine we are sitting in an outbreak-response meeting.

> *If this keeps going, how many cases are we looking at next month?*

We need a model before the situation outruns our intuition.

A bad first instinct

Here's what lab-confirmed case looked like from March 23.

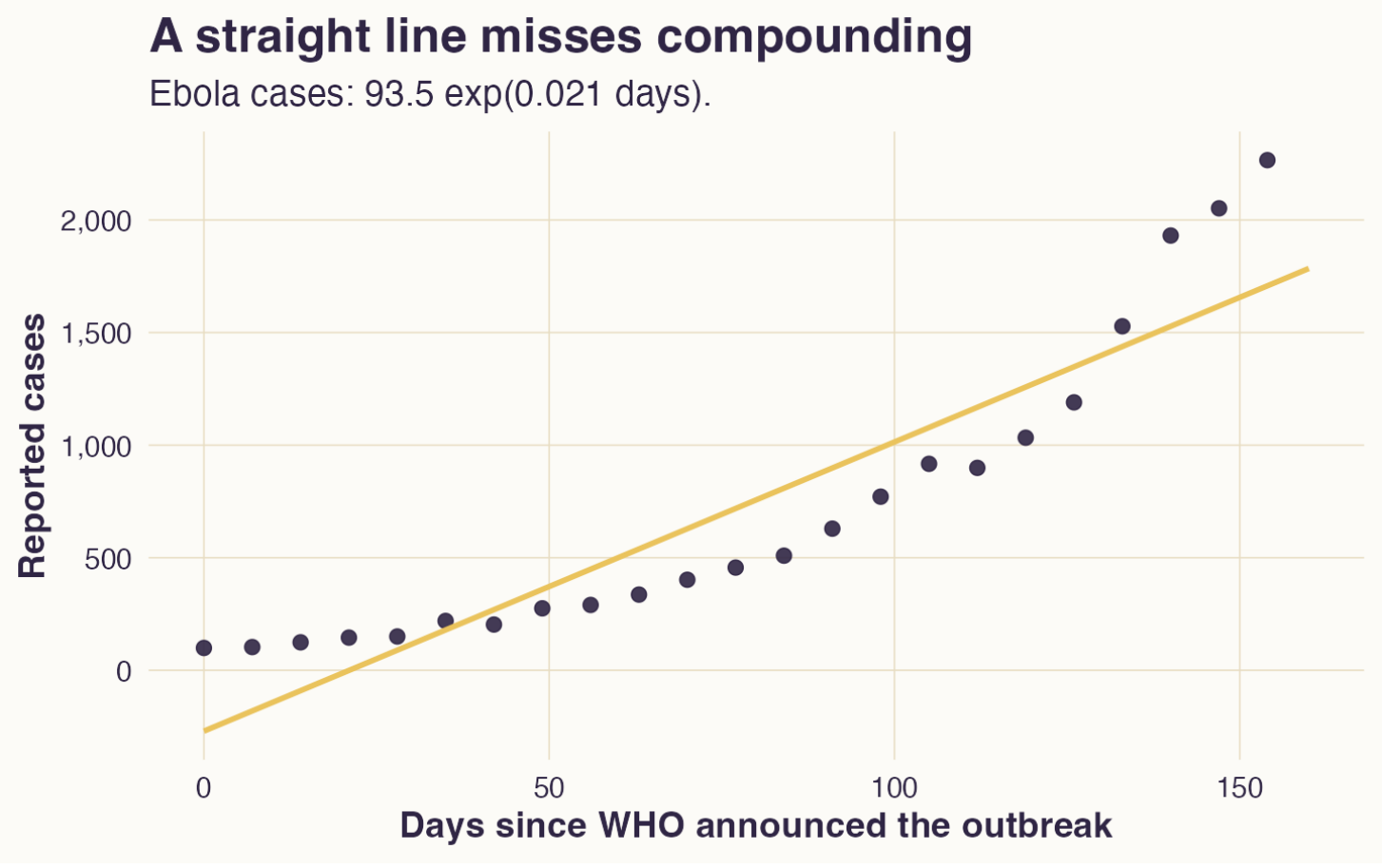


Figure 1: Scatterplot of Ebola cases over time since WHO announced the outbreak, with a straight-line fit that misses the compounding growth.

What the straight line says

A straight line forces one answer:

Each additional day adds the same number of cases.

That means:

- day 10 to day 11: add c cases,
- day 80 to day 81: add c cases,
- And so on.

But the outbreak is not adding the same number each day. It is compounding.

The model we want

For early epidemic growth, the natural model is:

> *Each day, the case count is multiplied by roughly the same factor.*

In this data set, it turns out that:

$$\frac{\text{cases tomorrow}}{\text{cases today}} \approx 1.021.$$

That is 2.1% growth per day. **How did we determine that?**

The exponential model

The exponential model expresses this as:

$$Y_t = Y_0 e^{rt}.$$

- Y_0: initial count at time $t = 0$.
- Y_t: expected count after t days.
- r: daily growth rate.
- t: days since the outbreak announcement.

For this example:

$$Y_t = 93.5 e^{0.021t}.$$

The curve fits the story

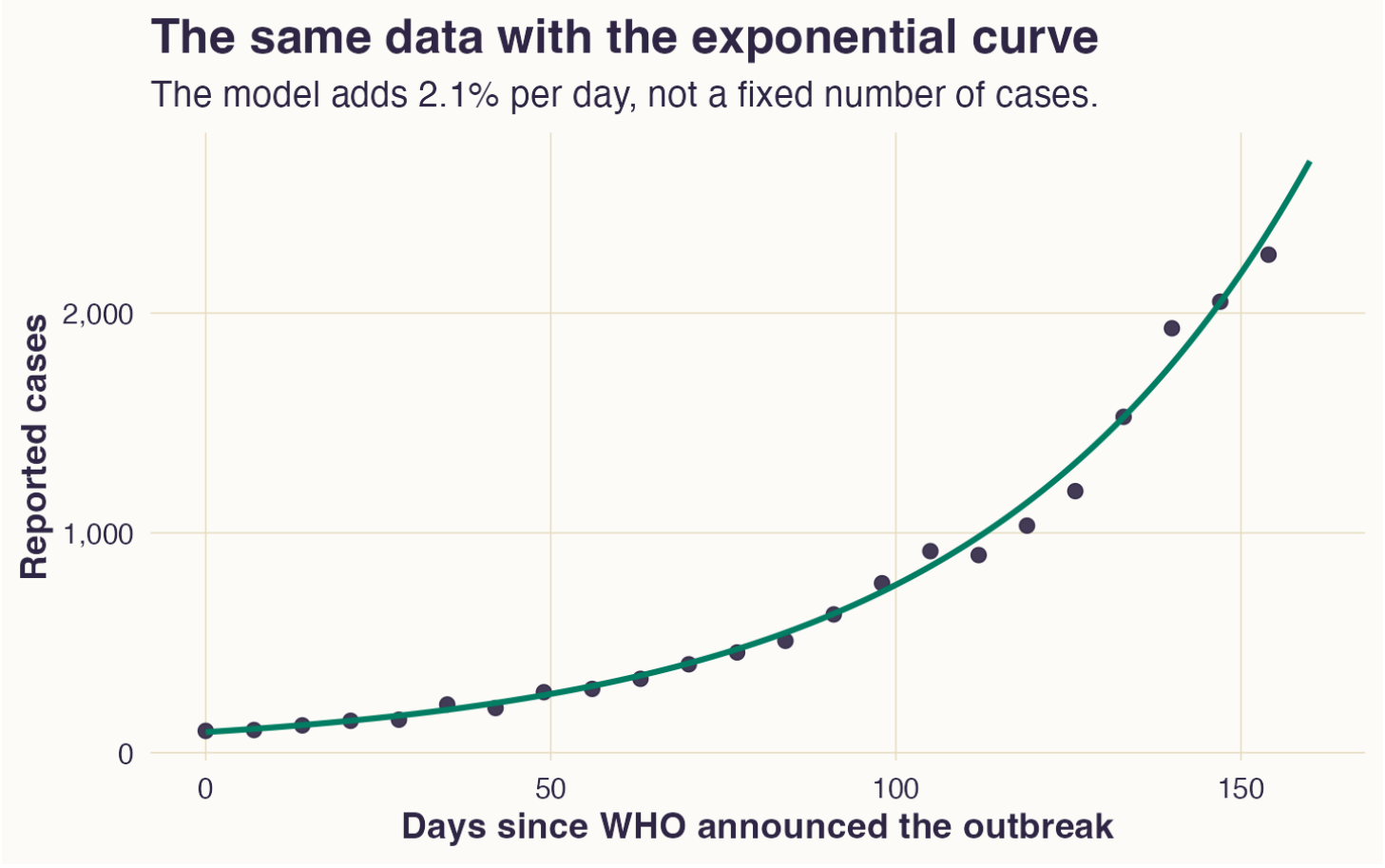


Figure 2: Scatterplot of Ebola cases by days since WHO announced the outbreak, with the fitted exponential curve.

So are we stuck?

No. Logs turn multiplication into addition:

$$\log(a \cdot b) = \log(a) + \log(b).$$

Start with the exponential model:

$$Y_t = Y_0 e^{rt}.$$

Take logs:

$$\log(Y_t) = \log(Y_0) + rt.$$

Now we have a straight-line model.

The same data on the log scale

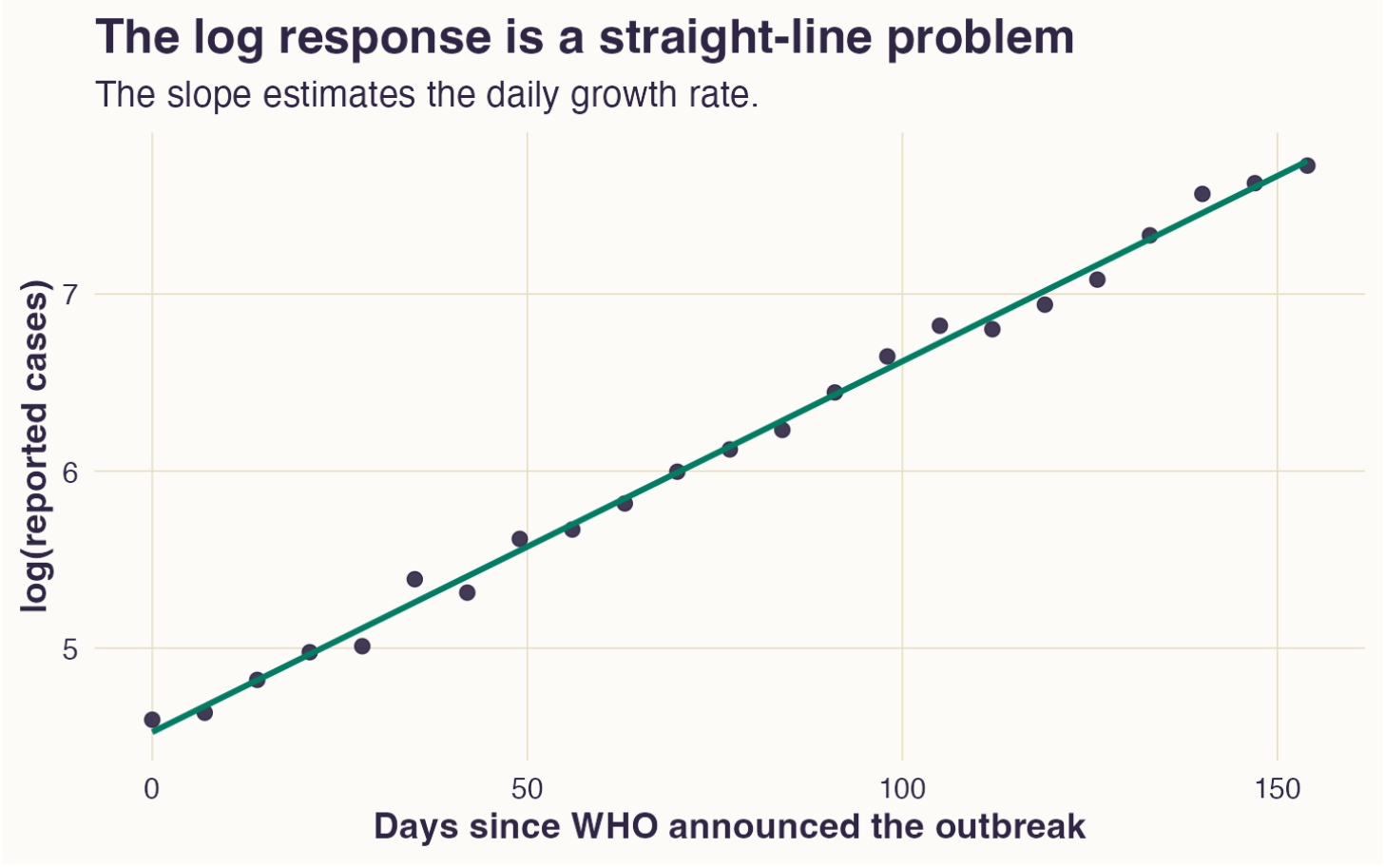


Figure 3: Scatterplot of log Ebola cases versus days since WHO announced the outbreak, with a fitted line.

What the fitted line means

On the log scale, the fitted equation is approximately:

$$\widehat{\log(\text{cases})} = 4.53 + 0.021 \cdot \text{day}.$$

Translation:

- 4.54 is the starting log count,
- 0.021 is the daily log-growth rate,
- $e^{0.021} \approx 1.021$: the fitted count multiplies by about 1.021 each day.

The slope is the percentage change each time period.

Comparing scales

On the log scale:

$$\widehat{\log(\text{cases})} = 4.53 + 0.021 \cdot \text{day}.$$

On the case-count scale:

$$\widehat{\text{cases}} = e^{4.53} e^{0.021 \cdot \text{day}} \approx 94 e^{0.021 \cdot \text{day}}.$$

Same model. The log scale is for fitting and interpretation. The original scale is where the cases live.

One day versus one month

A 2.1% daily growth rate sounds small.

For one day:

$$e^{0.021} \approx 1.021.$$

For 30 days:

$$e^{0.021 \cdot 30} \approx 1.88.$$

In words:

2.1% per day is about 88% growth over 30 days.

Doubling time

The doubling time solves for t:

$$2 = e^{rt}.$$

So:

$$t = \frac{\log(2)}{r}.$$

With $r = 0.021$:

$$\frac{\log(2)}{0.021} \approx 33 \text{ days.}$$

This is the number you would say in the meeting.

Rule of 70

A useful shortcut:

$$\text{doubling time} \approx \frac{70}{\text{growth rate as a percent}}.$$

For 2.1% daily growth:

$$70/2.1 \approx 33.$$

Just using the fact that $\log(2) \approx 0.70$.

What changed when we logged y?

We made a new response:

$$\log(y_i).$$

The regression is still a straight-line regression:

$$\log(y_i) = \beta_0 + \beta_1 x_i + \epsilon_i.$$

But the misses are no longer measured in units of y. They are measured in log units.

Same unit miss, different percent miss

Suppose a model predicts sales.

fitted sales	actual sales	unit residual	percent residual
100	120	+20	+20%
1000	1020	+20	+2%

A unit-scale model treats both misses as equally large.

But most businesses would not. Missing by 20 units is different when the baseline is 100 than when it is 1000.

Same percent miss, different unit miss

Now compare these:

fitted sales	actual sales	unit residual	ratio
100	120	+20	1.20
1000	1200	+200	1.20

A log-response model treats these as the same size miss.

Why?

Both actual values are 20% above the fitted value.

Log residuals are ratios

The log residual ϵ is:

$$\epsilon = \log(y) - \log(\widehat{y}).$$

Use the log rule:

$$\epsilon = \log\left(\frac{y}{\widehat{y}}\right).$$

So a log residual is a logged ratio.

If $\epsilon = \log(1.20)$, the actual value is 20% above fitted.

If $\epsilon = \log(0.80)$, the actual value is 20% below fitted.

What the error story becomes

Start with:

$$\log(y_i) = \beta_0 + \beta_1 x_i + \epsilon_i.$$

Exponentiate both sides:

$$y_i = e^{\beta_0 + \beta_1 x_i} \cdot e^{\epsilon_i}.$$

The fitted part sets the expected level.

The residual multiplies that level up or down.

When that error story makes sense

Log the response when typical misses feel like percentage misses.

- sales forecasts: missing by 10% feels comparable across stores of very different sizes,
- salary models: variation often grows with the salary level,
- house prices: price estimates for expensive houses usually have larger dollar misses,
- epidemic counts: uncertainty grows as the outbreak grows.

The log model changes what “a typical error” means.

A warning about zero

Logs require positive values.

- $\log(12.8)$ is fine.
- $\log(1)$ is fine.
- $\log(0.29)$ is fine.
- $\log(0)$ is undefined.
- $\log(-3)$ is not a real number.

Do not silently add 1, or 0.0001, because software complained.

Ask what zero means in the data.

Elasticity

Now suppose the predictor also lives on a ratio scale.

Question:

What happens to y when x changes by 1%?

That is no longer a unit-change question about x.

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Use a log-log model.

Example: body weight and brain weight

How does brain weight scale with body weight?

Adding 1 kg of body weight is not the same for every animal.

- Mouse to dog: big change from adding 20 kg.
- Horse to slightly larger horse: small change from adding 20 kg.

An additive slope is a strange target.

The original scale hides the pattern

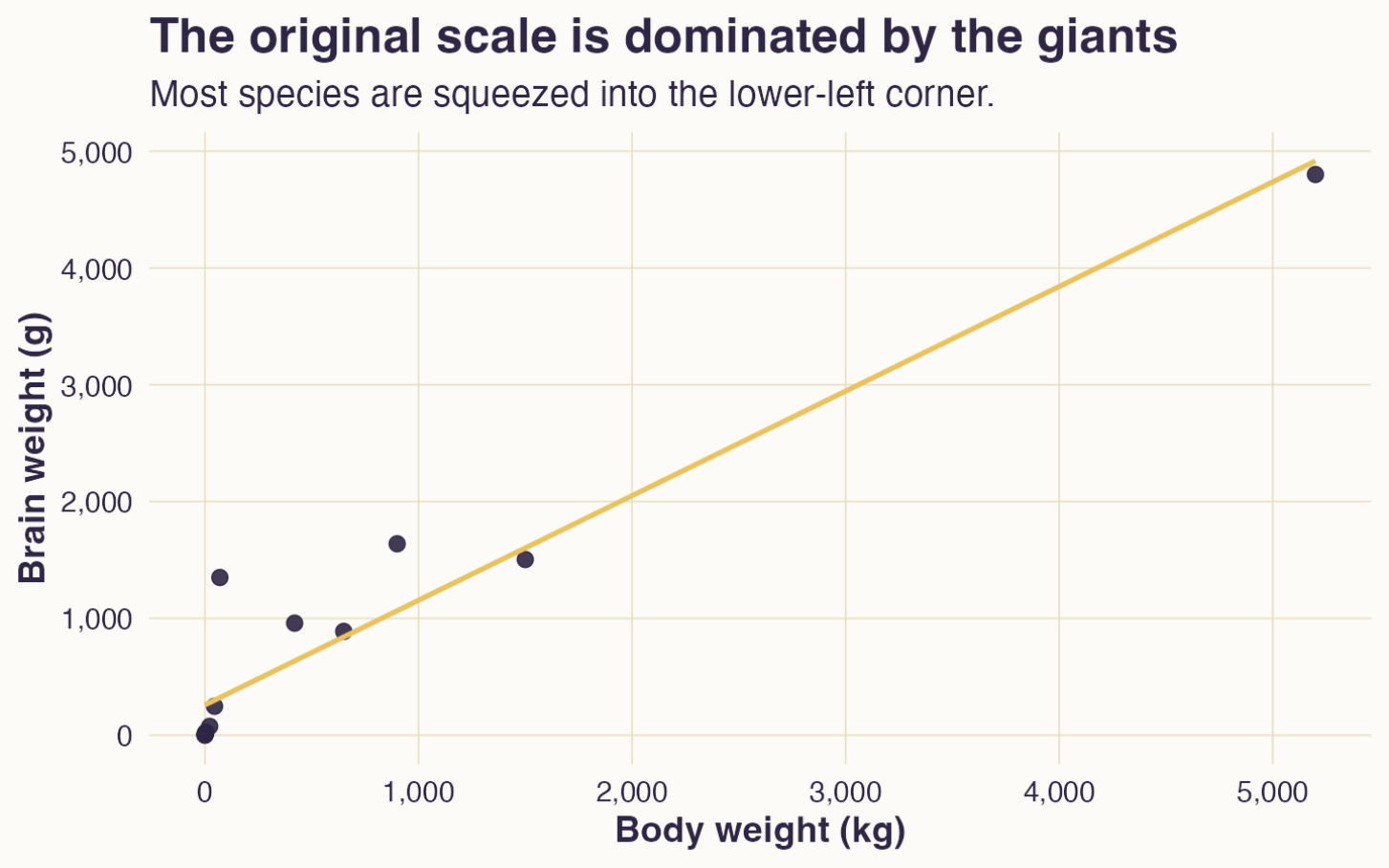


Figure 4: Scatterplot of mammal brain weight versus body weight on the original scale. A few large animals dominate the axes, hiding most of the data near the origin.

A power law

Many scaling relationships use:

$$y = Kx^{\beta}.$$

- x: body weight.
- y: brain weight.
- K: proportionality constant.
- β: scaling exponent.

If $\beta = 1$, brain weight scales proportionally with body weight.

If $\beta < 1$, brain weight grows more slowly than body weight.

The log-log trick

Take logs:

$$\log(y) = \log(K) + \beta \log(x).$$

For mammals:

$$\log(\text{brain}) = \alpha + \beta \log(\text{body}) + e.$$

This is a straight-line model for two logged variables.

The slope is an elasticity:

Elasticity

An elasticity answers:

If x changes by 1%, how many percent does y change?

In a log-log model, the elasticity is β.

$$\log(y) = \alpha + \beta \log(x),$$

The units disappear; percent talks to percent.

Elasticity

To see why, take derivatives of both sides:

$$\frac{d}{dx}\log(y) = \frac{d}{dx}\left[\alpha + \beta\log(x)\right].$$

Using the chain rule,

$$\frac{1}{y}\frac{dy}{dx} = \beta\frac{1}{x}.$$

Rearrange:

$$\beta = \frac{dy/y}{dx/x} = \frac{\%\text{ change in } y}{\%\text{ change in } x}$$

The log-log model for mammals

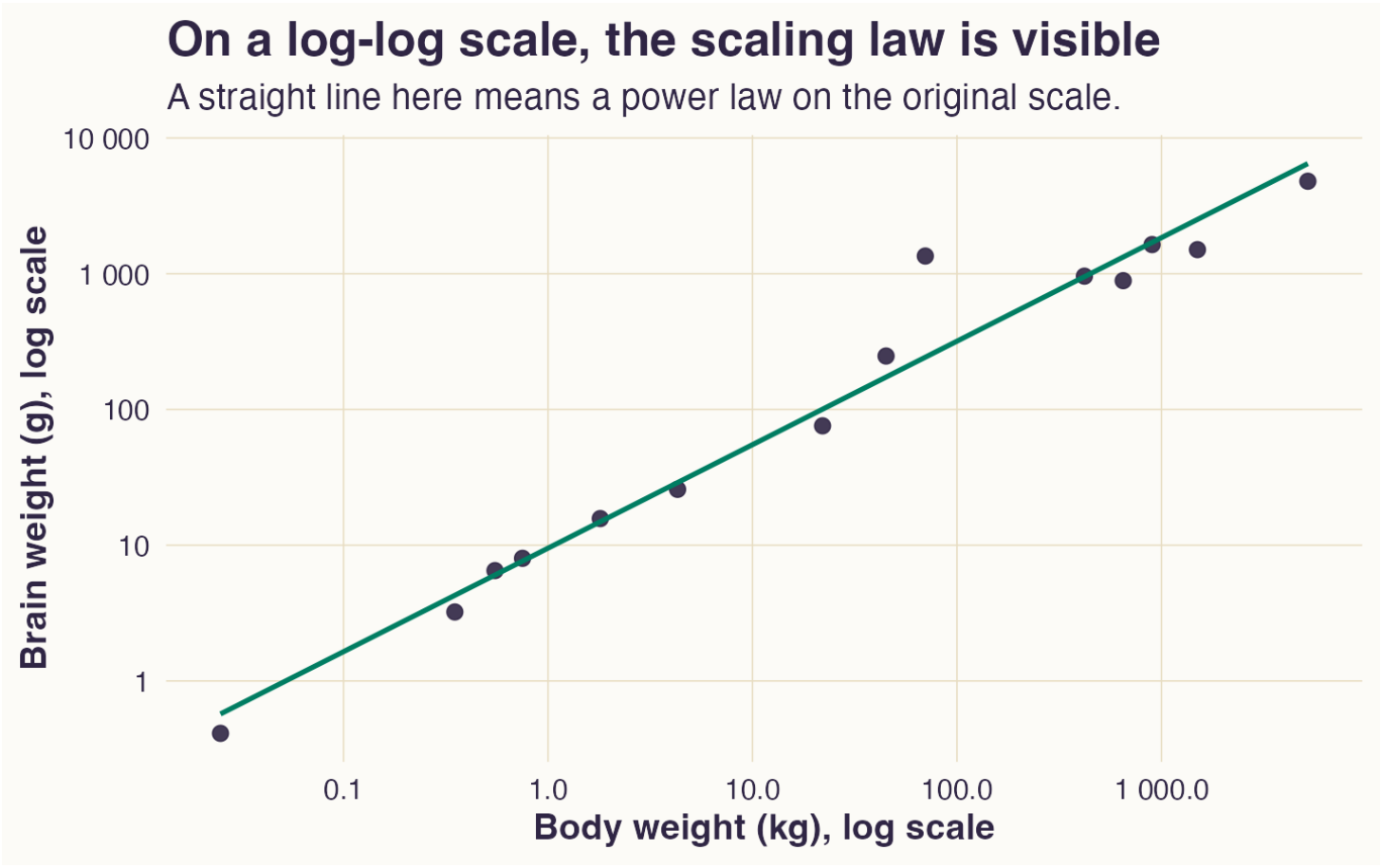


Figure 5: Scatterplot of log brain weight versus log body weight, with a fitted line. The species form an approximately straight increasing band.

Interpreting the mammal slope

Suppose the fitted equation is:

$$\widehat{\log(\text{brain})} = 2.25 + 0.76 \log(\text{body}).$$

The elasticity is 0.76.

In words:

A 1% increase in body weight corresponds to about a 0.76% increase in brain weight, on average.

Not grams.

Percent to percent.

A bigger move

If body weight doubles, predicted brain weight is multiplied by:

$$2^{0.76} \approx 1.69.$$

So doubling body weight predicts about 69% more brain weight.

Because $0.76 < 1$, brain weight grows less than proportionally with body weight.

Elasticities in economics

Examples:

- Price elasticity of demand: if price rises 1%, how much does quantity demanded fall?
- Income elasticity: if income rises 1%, how much does spending rise?
- Cross-price elasticity: if a competitor's price rises 1%, what happens to your sales?
- Wage elasticity of labor supply: if wages rise 1%, how much do hours worked change?
- Advertising elasticity: if ad spend rises 1%, how much do sales rise?

Same grammar, different economic question.

The online seller

An online seller runs a pricing experiment.

Visitors are randomized to see one price for the same product:

$$\$8, \$9, \$10, \ldots, \$32.$$

For each price, the seller records units sold.

The business question is:

If we raise price by 1%, how much demand do we lose?

That is a price-elasticity question.

Demand on the original scale

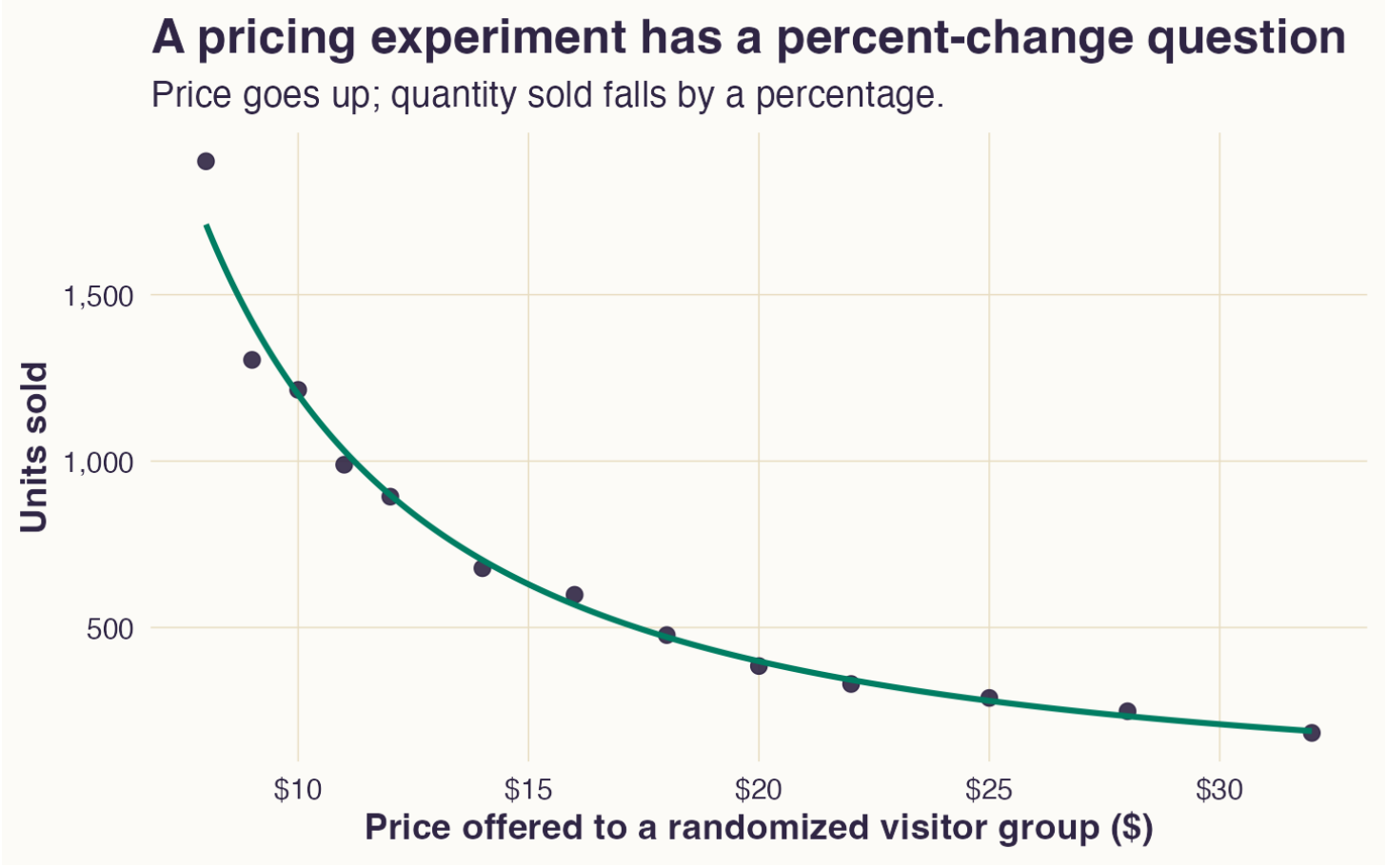


Figure 6: Scatterplot of units sold versus randomized price, with a downward-sloping fitted demand curve.

Demand on the log-log scale

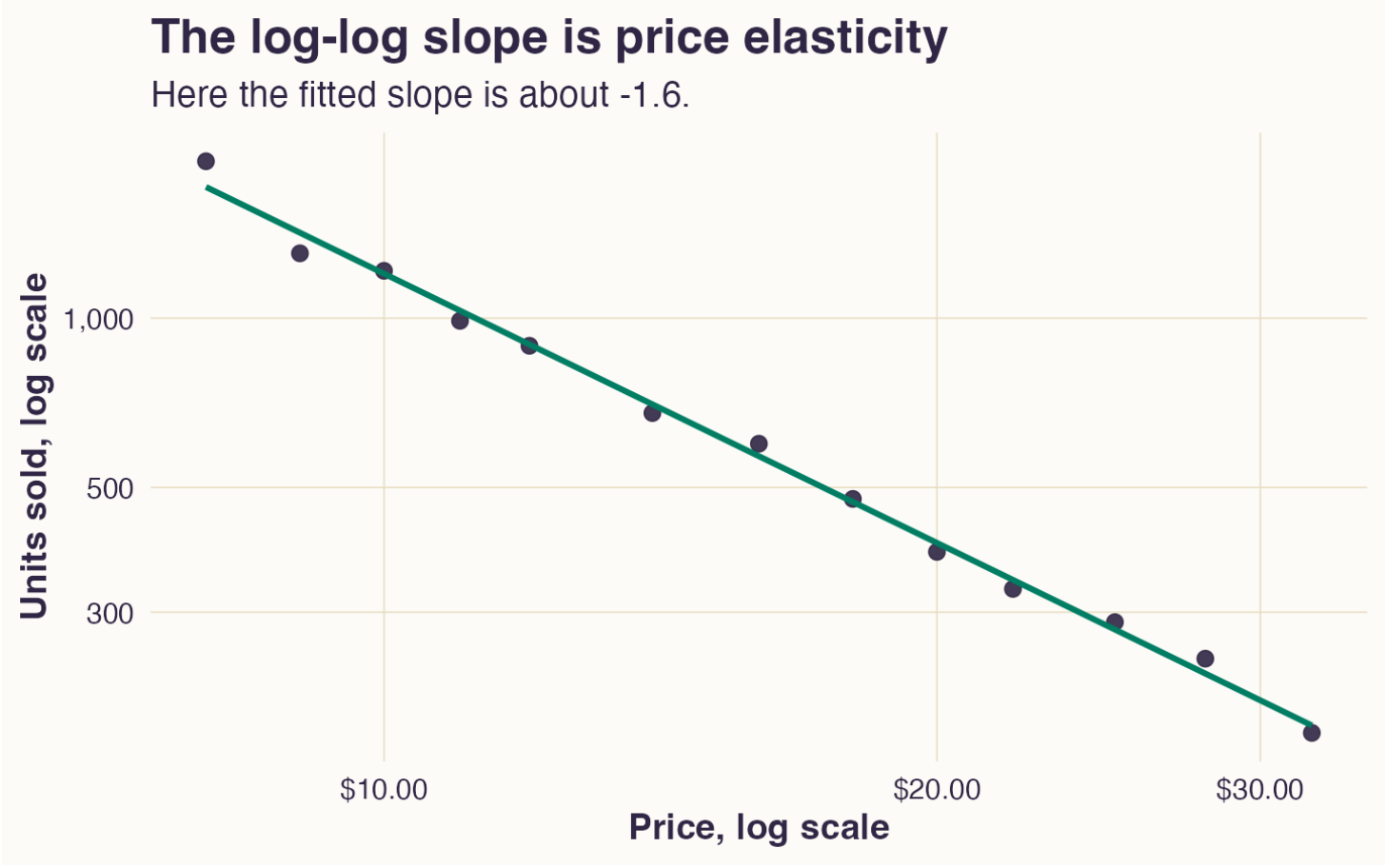


Figure 7: Log-log scatterplot of units sold versus randomized price, with a fitted line.

Reading the price elasticity

Suppose the fitted equation is:

$$\widehat{\log(\text{quantity})} = 10.75 - 1.59 \log(\text{price}).$$

The slope is -1.59.

In words:

> *A 1% increase in price predicts a 1.6% decrease in quantity sold.*

That is elastic demand: quantity moves more than price, in percent terms.

Why the sign matters

Price elasticity is usually negative: higher price, lower quantity.

If the slope were -0.3:

- a 1% price increase predicts a 0.3% quantity decrease,
- demand is inelastic,
- typically revenue tends to rise when price rises.

If the slope were -1.6:

- a 1% price increase predicts a 1.6% quantity decrease,
- demand is elastic,
- typically revenue tends to fall when price rises.

Why revenue changes

Revenue is:

$$\text{revenue} = \text{price} \times \text{quantity}.$$

If price rises 1% and quantity falls 1.6%, then revenue changes by roughly:

$$1\% - 1.6\% = -0.6\%.$$

Not exact calculus, but still a good first read.

Three common log models

Model	Fit this	Slope means
level-level	y on x	additive change in y per unit of x
log-level	$\log(y)$ on x	percent change in y per unit of x
log-log	$\log(y)$ on $\log(x)$	percent change in y per percent change in x

The model form tells you what comparison you are making.

A decision rule

Use logs when the question is naturally about multiplication, ratios, or percentages.

- growth happens by compounding,
- changes are reported as rates,
- the spread grows with the level,
- the original-scale plot is dominated by large values,
- on a log-scale plot, the data look like they follow a linear relationship.

This is a modeling choice, driven by a combination of data exploration and prior knowledge.

What can go wrong?

The common mistakes:

- interpreting a log-scale slope as a unit-change slope,
- fitting on logs and never translating back,
- using logs to hide outliers,
- forgetting that errors are now ratio errors,
- adding 1 to handle zeros without thinking.

The transformation is useful, but the interpretation has to follow it.

The practical lesson

Logs are useful because they turn some nonlinear stories into straight-line stories:

$$\text{compounding} \rightarrow \text{log-response line}$$

$$\text{power law} \rightarrow \text{log-log line}$$

But logging a variable changes the question.

- Level models speak in units.
- Log models speak in percentages and ratios.